
:Customer Journey Map for the Human Development Index (HDI) Prediction System

Stage 1 – Data Collection

Actions

- Collect country development data.
- Enter Life Expectancy.
- Enter Mean Years of Schooling.
- Enter Expected Years of Schooling.
- Enter Gross National Income (GNI) per capita.
- Upload or enter the dataset.

Touchpoints

- Web Application
- Data Input Form
- Dataset Upload Interface

Customer Thoughts

- Is my data accurate?
- Have I entered all required values?
- Will the prediction be reliable?

Customer Feelings

- Curious
- Hopeful
- Interested

Process Ownership

- Data Analyst
- Research Team

- User

Improvement Opportunities

- Automatic data validation
 - Real-time dataset updates
 - User-friendly input interface
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Stage 2 – HDI Prediction

Actions

- Click the **Predict** button.
- Machine Learning model processes the input.
- System predicts the HDI category.

Touchpoints

- Prediction Engine
- Machine Learning Model
- Processing Dashboard

Customer Thoughts

- How is the prediction calculated?
- Which indicator has the highest impact?
- Is the prediction accurate?

Customer Feelings

- Excited
- Confident
- Expectant

Process Ownership

- Machine Learning System
- AI Prediction Model
- Backend Server

Improvement Opportunities

- Improve prediction accuracy
- Faster processing
- Explainable AI for transparency

Stage 3 – Result Analysis & Decision Making

Actions

- View predicted HDI score.
- Analyze the predicted category.
- Compare development indicators.
- Use insights for policy planning.

Touchpoints

- Prediction Dashboard
- Charts & Graphs
- Report Generation Module

Customer Thoughts

- Which sectors require improvement?
- What policies should be prioritized?
- How can these insights improve national development?

Customer Feelings

- Satisfied
- Informed
- Motivated

Process Ownership

- Government Officials
- Policy Makers
- Researchers

Improvement Opportunities

- Interactive dashboards
- Personalized recommendations
- Export reports in PDF and Excel
- Historical trend analysis

Summary

The Customer Journey Map demonstrates how users interact with the **HDI Prediction System** from entering development indicators to receiving predictive insights. It identifies user needs, expectations, and pain points while highlighting opportunities to enhance usability, prediction accuracy, and decision support. This approach ensures that the proposed machine learning system delivers meaningful insights for governments, researchers, educators, and international development organizations.

